

# HAORAN JIN

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## EDUCATION

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**University of Michigan**

Sep. 2023 - present

**Ph.D Candidate** in Computer Engineering | Computer Architecture & EDA

Advisor: Prof. Nathaniel Bleier & Previous Advisor: Prof. George Tzimpragos

**Shanghai Jiao Tong University**

Sep. 2019 - Aug. 2023

**B.Eng** in Electrical and Computer Engineering

## HONORS AND AWARDS

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- 2026 Rackham Conference Travel Grant
- 2019,2020,2021,2022 John Wu & Jane Sun Sunshine Scholarship
- 2020,2021,2022 SJTU Undergraduate Excellence Scholarship

## PUBLICATION

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- Haoran Jin\*, Kangqi Zhang\*, Jirong Yang, Barry Lyu, Qiuyi Ding, Ruijie Gao, Nathan Bleier, *Precision As You Need: Stochastic Computing Is a Dense Adaptive Quantizer*, NeurIPS 2026 (Under Review)
- Ruijie Gao, Haoran Jin, Jirong Yang, Nathaniel Bleier, *ATLAAS: Automatic Tensor-Level Abstraction of Accelerator Semantics*, arXiv 2026
- Haoran Jin, Jirong Yang, Zhiheng Zhang, Justin Shin, Barry Lyu, Kangqi Zhang, Yunpeng Liu, Nathaniel Bleier, *Fengshui: Demystifying Chiplet Ecosystem and Bespoke Neural Network Accelerator Codesign*, MICRO 2026 (Under Review)
- Haoran Jin\*, Jirong Yang\*, Barry Lyu, Ruijie Gao, Nathaniel Bleier, *æSIP:  $\mu$ Arch-aware ASIP-ISA Co-Design via Program Synthesis, Equality Saturation, and External Don't Cares*, ISCA 2026
- Panagiotis Papanikolaou, Alex Vanasse, Haoran Jin, Georgios Tzimpragos, Jennifer Volk, *Duplication-Aware Retiming and Cell Interface Redesign for Superconductor Circuit Minimization*, DAC 2026
- Zexi Li\*, Haoran Jin\*, Kuncai Zhong, Guojie Luo, Runsheng Wang, Weikang Qian, *SCGen: A Versatile Generator Framework for Agile Development of Stochastic Circuits*, DATE 2024
- Kuncai Zhong, Zexi Li, Haoran Jin, Weikang Qian, *Exploiting Uniform Spatial Distribution to Design Efficient Random Number Source for Stochastic Computing*, ICCAD 2022

## WORK EXPERIENCE

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Intern, Architecture at *Tenstorrent*

May. 2026 - present

- Build performance model and do performance correlation for AI hardware

Graduate Student Instructor at *University of Michigan*

Jan. 2026 – May. 2026

- Organized the course, EECS 598 Computer Architecture for Post-Moore Era

## SELECTED PROJECTS

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**æSIP:  $\mu$ Arch-aware ASIP-ISA Co-Design via Program Synthesis, Equality Saturation, and External Don't Cares**

Prof. Nathaniel Bleier at *University of Michigan*

July. 2025 – Nov. 2025

- Proposed æSIP, the first  $\mu$ Arch-aware ASIP-ISA co-design framework that leverages hardware aware program rewriting to automatically generate an optimized ASIP-program pair.

- Built a program synthesis framework that leverages neuro-symbolic approach to automatically discover correctness-preserving rewrites that maximize hardware removal opportunities.

### **Mozart: An Ecosystem-Accelerator Codesign Framework for Composable, Heterogeneous Chiplet-based Neural Network Accelerators**

Prof. Nathaniel Bleier at *University of Michigan*

Jan. 2025 – Aug. 2025

- Developed a chiplet ecosystem-accelerator codesign framework to increase chiplet reuse and hence reducing non-recurring engineering costs.
- Built hierarchical design space exploration methodology to systematically compose heterogeneous chiplet-based bespoke ASIC leveraging operator-level disaggregation insights.

### **Minimum Area Retiming for xSFQ logic**

Prof. George Tzimpragos at *University of Michigan*

Aug. 2024 – Dec. 2024

- Developed a linear programming formulation to simultaneously minimize the number of registers and duplication logic in xSFQ circuits for optimal area reduction while maintaining clock period constraints.

### **Mixed Circuit of Binary & Stochastic Computing**

Prof. Weikang Qian at *SJTU*

Jan. 2022 – Sep. 2023

- Defined a domain-specific language (DSL) for agile mixed circuit development.
- Evaluated mixed binary-stochastic circuits for image processing applications.

## **SKILL**

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### **Programing Tools**

C, C++, Python, Java, Rust, Verilog, Chisel, ELM, Racket  
Triton, Timeloop, MatLab, Mathematica, Vitis HLS, Spark, Egglog, Hadoop, Pin, Pspice